Motion

The big idea

A body is in **motion** if its position changes with time **relative to a reference point**. To describe motion precisely we need distance, displacement, speed, velocity and acceleration, and graphs and equations that connect them.

Distance and displacement

- **Distance**: total path length; a **scalar** (magnitude only).
- **Displacement**: shortest distance from the initial to the final position, **with direction**; a **vector**. It can be zero even when distance isn't (a full lap of a circular track).

Speed, velocity and acceleration

RATES OF MOTION

Average speed = total distance / total time

Velocity = displacement / time (speed in a given direction)

For uniform acceleration, average velocity = (u + v)/2

Acceleration a = (v − u)/t (unit m/s²)

Uniform motion: equal distances in equal intervals. **Non-uniform**: unequal. A negative acceleration (slowing down) is called **retardation** or deceleration.

Graphs of motion

- **Distance–time graph**: slope = **speed**. A straight sloping line = uniform speed; a curve = changing speed; a horizontal line = at rest.
- **Velocity–time graph**: slope = **acceleration**; **area under the graph = displacement**.

Equations of motion (uniform acceleration)

THE THREE EQUATIONS

$$v = u + at$$

$$s = ut + \tfrac{1}{2}at^2$$

$$v^2 - u^2 = 2as$$

A CAR ACCELERATES UNIFORMLY FROM 18 KM/H TO 36 KM/H IN 5 S. FIND A AND THE DISTANCE COVERED.

Convert: u = 5 m/s, v = 10 m/s. a = (10 − 5)/5 = **1 m/s²**.

s = ut + ½at² = 25 + 12.5 = **37.5 m**.

UNIT CONVERSION

To change km/h to m/s, multiply by 5/18. 72 km/h = 20 m/s.

Uniform circular motion

A body moving in a circle at **constant speed** is still **accelerating**, because its **direction** keeps changing. Speed v = 2πr / t. Examples: the Moon around Earth, an athlete on a circular track, a stone whirled on a string.

CONSTANT SPEED ≠ CONSTANT VELOCITY

In circular motion the speed is constant but the velocity is not, because velocity includes direction.

Quick recap

- Distance is scalar; displacement is vector.
- Slope of x–t = speed; slope of v–t = acceleration; area under v–t = displacement.
- $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 - u^2 = 2as$.
- Uniform circular motion is accelerated motion.

Practice questions

1. An athlete completes one round of a circular track. The displacement is
   (a) zero (b) equal to the circumference (c) equal to the diameter (d) infinite
2. The slope of a velocity–time graph gives
   (a) distance (b) speed (c) acceleration (d) displacement
3. 54 km/h in m/s is
   (a) 54 (b) 15 (c) 20 (d) 10
4. A bus starting from rest reaches 20 m/s in 10 s. Its acceleration is
   (a) 0.5 m/s² (b) 2 m/s² (c) 200 m/s² (d) 10 m/s²
5. Uniform circular motion is accelerated because
   (a) speed changes (b) direction of velocity changes (c) the body stops (d) distance is zero
6. The area under a velocity–time graph gives
   (a) acceleration (b) displacement (c) speed (d) time

Answers: 1. a 2. c 3. b 4. b 5. b 6. b